



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science and Engineering

"A place to learn; A chance to grow"

2025-26 (Even Semester)



MSE-1 EXAMINATION

Course: B.Tech. In Computer Science and Engineering

Year/Semester: 6th Semester (3rd Year)

Subject: Robotic Automation System

Subject Code: CS6E004D

Max Marks: 20

Date: 07/02/2026

Duration: 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

- CO1: Students will Understand the fundamental principles of robotics and industrial automation.
CO2: Students will Select and apply appropriate sensors, actuators, and control techniques.
CO3: Students will analyse AVR RISC Microcontroller architecture.
CO4: Students will Develop and simulate ARM Processor.
CO5: Students will be able to Evaluate real-world robotic automation systems and current trends.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Embedded System with an example.	L1/CO1	1
Q.2	Classify the types of Robots.	L2/CO2	1
Q.3	State the application Cartesian Robots.	L1/CO2	1
Q.4	List the characteristics of Embedded system.	L2/CO1	1

SECTION B

1. All Questions are compulsory.
2. Answer in one two three sentences.

Q.5	List the Hardware used in the Architecture of Embedded System.	L1/CO1	2
Q.6	Enlist the advantages & disadvantages of Cartesian Robot.	L1/CO2	2
Q.7	Differentiate between Microprocessor and Microcontroller.	L4/CO1	2
Q.8	Define Robots and list the components used in Robot.	L1/CO1	2

SECTION C

1. Solve any TWO questions.

Q.9	Explain the Categories of Embedded Systems.	L2/CO1	4
Q.10	Illustrate the Recent Trends in Embedded Systems.	L3/CO1	4
Q.11	Describe Cartesian Robot with their example.	L2/CO2	4